

Newman-Penrose Formalism Calculations

The Wahlquist Metric

Metric

$$ds^2 = \left(\frac{U(z) - V(\omega)}{v_1(z) + v_2(\omega)} \right) d\tau \otimes d\tau + \left(\frac{V(\omega) v_1(z) + U(z) v_2(\omega)}{v_1(z) + v_2(\omega)} \right) d\tau \otimes d\sigma + \left(\frac{V(\omega) v_1(z) + U(z) v_2(\omega)}{v_1(z) + v_2(\omega)} \right) d\sigma \otimes d\tau \\ + \left(-\frac{V(\omega) v_1(z)^2 - U(z) v_2(\omega)^2}{v_1(z) + v_2(\omega)} \right) d\sigma \otimes d\sigma + \left(\frac{v_1(z) + v_2(\omega)}{V(\omega)} \right) d\omega \otimes d\omega + \left(\frac{v_1(z) + v_2(\omega)}{U(z)} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (\tau, \sigma, \omega, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \sqrt{\frac{V(\omega)}{v_1(z) + v_2(\omega)}} \right) d\tau + \left(-\frac{1}{2} \sqrt{2} \sqrt{\frac{V(\omega)}{v_1(z) + v_2(\omega)}} v_1(z) \right) d\sigma + \left(\frac{\sqrt{2}}{2 \sqrt{\frac{V(\omega)}{v_1(z) + v_2(\omega)}}} \right) d\omega$$

$$n = \left(\frac{1}{2} \sqrt{2} \sqrt{\frac{V(\omega)}{v_1(z) + v_2(\omega)}} \right) d\tau + \left(-\frac{1}{2} \sqrt{2} \sqrt{\frac{V(\omega)}{v_1(z) + v_2(\omega)}} v_1(z) \right) d\sigma + \left(-\frac{\sqrt{2}}{2 \sqrt{\frac{V(\omega)}{v_1(z) + v_2(\omega)}}} \right) d\omega$$

$$m = \left(\frac{1}{2} \sqrt{2} \sqrt{\frac{U(z)}{v_1(z) + v_2(\omega)}} \right) d\tau + \left(\frac{1}{2} \sqrt{2} \sqrt{\frac{U(z)}{v_1(z) + v_2(\omega)}} v_2(\omega) \right) d\sigma + \left(\frac{i \sqrt{2}}{2 \sqrt{\frac{U(z)}{v_1(z) + v_2(\omega)}}} \right) dz$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \sqrt{\frac{U(z)}{v_1(z) + v_2(\omega)}} \right) d\tau + \left(\frac{1}{2} \sqrt{2} \sqrt{\frac{U(z)}{v_1(z) + v_2(\omega)}} v_2(\omega) \right) d\sigma + \left(-\frac{i \sqrt{2}}{2 \sqrt{\frac{U(z)}{v_1(z) + v_2(\omega)}}} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{v_1(z) + v_2(\omega)}v_2(\omega)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{V(\omega)}} \right) d\tau + \left(\frac{\sqrt{v_1(z) + v_2(\omega)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{V(\omega)}} \right) d\sigma + \left(\frac{\sqrt{v_1(z) + v_2(\omega)}\sqrt{V(\omega)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega)} \right) d\omega$$

$$n = \left(-\frac{\sqrt{v_1(z) + v_2(\omega)}v_2(\omega)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{V(\omega)}} \right) d\tau + \left(\frac{\sqrt{v_1(z) + v_2(\omega)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{V(\omega)}} \right) d\sigma + \left(-\frac{\sqrt{v_1(z) + v_2(\omega)}\sqrt{V(\omega)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega)} \right) d\omega$$

$$m = \left(\frac{\sqrt{v_1(z) + v_2(\omega)}v_1(z)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{U(z)}} \right) d\tau + \left(\frac{\sqrt{v_1(z) + v_2(\omega)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{U(z)}} \right) d\sigma + \left(\frac{i\sqrt{v_1(z) + v_2(\omega)}\sqrt{U(z)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega)} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{v_1(z) + v_2(\omega)}v_1(z)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{U(z)}} \right) d\tau + \left(\frac{\sqrt{v_1(z) + v_2(\omega)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{U(z)}} \right) d\sigma + \left(-\frac{i\sqrt{v_1(z) + v_2(\omega)}\sqrt{U(z)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega)} \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{v_1(z) + v_2(\omega)}v_2(\omega)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{V(\omega)}} \partial_\tau + \frac{\sqrt{v_1(z) + v_2(\omega)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{V(\omega)}} \partial_\sigma + \frac{\sqrt{v_1(z) + v_2(\omega)}\sqrt{V(\omega)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega)} \partial_\omega$$

$$\Delta = -\frac{\sqrt{v_1(z) + v_2(\omega)}v_2(\omega)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{V(\omega)}} \partial_\tau + \frac{\sqrt{v_1(z) + v_2(\omega)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{V(\omega)}} \partial_\sigma - \frac{\sqrt{v_1(z) + v_2(\omega)}\sqrt{V(\omega)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega)} \partial_\omega$$

$$\delta = \frac{\sqrt{v_1(z) + v_2(\omega)}v_1(z)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{U(z)}} \partial_\tau + \frac{\sqrt{v_1(z) + v_2(\omega)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{U(z)}} \partial_\sigma + \frac{i\sqrt{v_1(z) + v_2(\omega)}\sqrt{U(z)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega)} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{v_1(z) + v_2(\omega)}v_1(z)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{U(z)}} \partial_\tau + \frac{\sqrt{v_1(z) + v_2(\omega)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega))\sqrt{U(z)}} \partial_\sigma - \frac{i\sqrt{v_1(z) + v_2(\omega)}\sqrt{U(z)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(\omega)} \partial_z$$

Spin Coefficients

$$\rho = -\frac{i\sqrt{2}\sqrt{v_1(z)+v_2(\omega)}\sqrt{V(\omega)}\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^2+2v_1(z)v_2(\omega)+v_2(\omega)^2\right)} - \frac{\sqrt{2}\sqrt{v_1(z)+v_2(\omega)}\sqrt{V(\omega)}\frac{\partial}{\partial \omega}v_2(\omega)}{4\left(v_1(z)^2+2v_1(z)v_2(\omega)+v_2(\omega)^2\right)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{i\sqrt{2}\sqrt{v_1(z)+v_2(\omega)}\sqrt{V(\omega)}\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^2+2v_1(z)v_2(\omega)+v_2(\omega)^2\right)} - \frac{\sqrt{2}\sqrt{v_1(z)+v_2(\omega)}\sqrt{V(\omega)}\frac{\partial}{\partial \omega}v_2(\omega)}{4\left(v_1(z)^2+2v_1(z)v_2(\omega)+v_2(\omega)^2\right)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{i\sqrt{v_1(z)+v_2(\omega)}v_1(z)\frac{\partial}{\partial z}U(z)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)\sqrt{U(z)}} + \frac{i\sqrt{v_1(z)+v_2(\omega)}v_2(\omega)\frac{\partial}{\partial z}U(z)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)\sqrt{U(z)}} \\ - \frac{i\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial z}v_1(z)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} - \frac{\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial \omega}v_2(\omega)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)}$$

$$\beta = \frac{i\sqrt{v_1(z)+v_2(\omega)}v_1(z)\frac{\partial}{\partial z}U(z)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)\sqrt{U(z)}} + \frac{i\sqrt{v_1(z)+v_2(\omega)}v_2(\omega)\frac{\partial}{\partial z}U(z)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)\sqrt{U(z)}} \\ - \frac{i\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial z}v_1(z)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} - \frac{\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial \omega}v_2(\omega)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)}$$

$$\tau = -\frac{i\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial z}v_1(z)}{2\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} - \frac{\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial \omega}v_2(\omega)}{2\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)}$$

$$\begin{aligned}
\pi &= -\frac{i\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial z}v_1(z)}{2\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} - \frac{\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial\omega}v_2(\omega)}{2\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} \\
\epsilon &= \frac{\sqrt{v_1(z)+v_2(\omega)}v_1(z)\frac{\partial}{\partial\omega}V(\omega)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)\sqrt{V(\omega)}} + \frac{\sqrt{v_1(z)+v_2(\omega)}v_2(\omega)\frac{\partial}{\partial\omega}V(\omega)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)\sqrt{V(\omega)}} \\
&\quad - \frac{i\sqrt{v_1(z)+v_2(\omega)}\sqrt{V(\omega)}\frac{\partial}{\partial z}v_1(z)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} - \frac{\sqrt{v_1(z)+v_2(\omega)}\sqrt{V(\omega)}\frac{\partial}{\partial\omega}v_2(\omega)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} \\
\gamma &= \frac{\sqrt{v_1(z)+v_2(\omega)}v_1(z)\frac{\partial}{\partial\omega}V(\omega)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)\sqrt{V(\omega)}} + \frac{\sqrt{v_1(z)+v_2(\omega)}v_2(\omega)\frac{\partial}{\partial\omega}V(\omega)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)\sqrt{V(\omega)}} \\
&\quad - \frac{i\sqrt{v_1(z)+v_2(\omega)}\sqrt{V(\omega)}\frac{\partial}{\partial z}v_1(z)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} - \frac{\sqrt{v_1(z)+v_2(\omega)}\sqrt{V(\omega)}\frac{\partial}{\partial\omega}v_2(\omega)}{4\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)}
\end{aligned}$$

Spin Coefficients Relations

$$\begin{aligned}
\bar{\alpha} + \beta &= -\frac{\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial\omega}v_2(\omega)}{2\left(\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2\right)} \\
\bar{\tau} + \pi &= -\frac{\sqrt{v_1(z)+v_2(\omega)}\sqrt{U(z)}\frac{\partial}{\partial\omega}v_2(\omega)}{\sqrt{2}v_1(z)^2+2\sqrt{2}v_1(z)v_2(\omega)+\sqrt{2}v_2(\omega)^2}
\end{aligned}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} &= \frac{V(\omega)\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} + \frac{V(\omega)\frac{\partial}{\partial\omega}v_2(\omega)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\
&\quad - \frac{V(\omega)v_1(z)\frac{\partial^2}{(\partial\omega)^2}v_2(\omega)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{V(\omega)v_2(\omega)\frac{\partial^2}{(\partial\omega)^2}v_2(\omega)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)}
\end{aligned}$$

$$\Phi_{01} = \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^2 + 2v_1(z)v_2(\omega) + v_2(\omega)^2\right)} - \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{8\left(v_1(z)^2 + 2v_1(z)v_2(\omega) + v_2(\omega)^2\right)}$$

$$\begin{aligned}\Phi_{02} = & -\frac{U(z)\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} + \frac{U(z)v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} \\ & + \frac{U(z)v_2(\omega)\frac{\partial^2}{(\partial z)^2}v_1(z)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} - \frac{U(z)\frac{\partial}{\partial \omega}v_2(\omega)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)}\end{aligned}$$

$$\Phi_{10} = \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^2 + 2v_1(z)v_2(\omega) + v_2(\omega)^2\right)} - \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{8\left(v_1(z)^2 + 2v_1(z)v_2(\omega) + v_2(\omega)^2\right)}$$

$$\begin{aligned}
\Phi_{11} = & -\frac{v_1(z)^2 \frac{\partial^2}{(\partial z)^2} U(z)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} - \frac{v_1(z) v_2(\omega) \frac{\partial^2}{(\partial z)^2} U(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& - \frac{v_2(\omega)^2 \frac{\partial^2}{(\partial z)^2} U(z)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} + \frac{v_1(z)^2 \frac{\partial^2}{(\partial \omega)^2} V(\omega)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& + \frac{v_1(z) v_2(\omega) \frac{\partial^2}{(\partial \omega)^2} V(\omega)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} + \frac{v_2(\omega)^2 \frac{\partial^2}{(\partial \omega)^2} V(\omega)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& + \frac{v_1(z) \frac{\partial}{\partial z} U(z) \frac{\partial}{\partial z} v_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} + \frac{v_2(\omega) \frac{\partial}{\partial z} U(z) \frac{\partial}{\partial z} v_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& - \frac{3U(z) \frac{\partial}{\partial z} v_1(z)^2}{16(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} + \frac{3V(\omega) \frac{\partial}{\partial z} v_1(z)^2}{16(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& + \frac{U(z) v_1(z) \frac{\partial^2}{(\partial z)^2} v_1(z)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} + \frac{U(z) v_2(\omega) \frac{\partial^2}{(\partial z)^2} v_1(z)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& - \frac{v_1(z) \frac{\partial}{\partial \omega} V(\omega) \frac{\partial}{\partial \omega} v_2(\omega)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} - \frac{v_2(\omega) \frac{\partial}{\partial \omega} V(\omega) \frac{\partial}{\partial \omega} v_2(\omega)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& - \frac{3U(z) \frac{\partial}{\partial \omega} v_2(\omega)^2}{16(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} + \frac{3V(\omega) \frac{\partial}{\partial \omega} v_2(\omega)^2}{16(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& - \frac{V(\omega) v_1(z) \frac{\partial^2}{(\partial \omega)^2} v_2(\omega)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} - \frac{V(\omega) v_2(\omega) \frac{\partial^2}{(\partial \omega)^2} v_2(\omega)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)}
\end{aligned}$$

$$\Phi_{12} = \frac{\sqrt{U(z)} \sqrt{V(\omega)} \frac{\partial^2}{(\partial z)^2} v_1(z)}{8(v_1(z)^2 + 2v_1(z) v_2(\omega) + v_2(\omega)^2)} - \frac{\sqrt{U(z)} \sqrt{V(\omega)} \frac{\partial^2}{(\partial \omega)^2} v_2(\omega)}{8(v_1(z)^2 + 2v_1(z) v_2(\omega) + v_2(\omega)^2)}$$

$$\begin{aligned}
\Phi_{20} = & -\frac{U(z) \frac{\partial}{\partial z} v_1(z)^2}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} + \frac{U(z) v_1(z) \frac{\partial^2}{(\partial z)^2} v_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} \\
& + \frac{U(z) v_2(\omega) \frac{\partial^2}{(\partial z)^2} v_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)} - \frac{U(z) \frac{\partial}{\partial \omega} v_2(\omega)^2}{8(v_1(z)^3 + 3v_1(z)^2 v_2(\omega) + 3v_1(z) v_2(\omega)^2 + v_2(\omega)^3)}
\end{aligned}$$

$$\Phi_{21} = \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^2 + 2v_1(z)v_2(\omega) + v_2(\omega)^2\right)} - \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{8\left(v_1(z)^2 + 2v_1(z)v_2(\omega) + v_2(\omega)^2\right)}$$

$$\begin{aligned}\Phi_{22} = & \frac{V(\omega)\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} + \frac{V(\omega)\frac{\partial}{\partial \omega}v_2(\omega)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} \\ & - \frac{V(\omega)v_1(z)\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} - \frac{V(\omega)v_2(\omega)\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)}\end{aligned}$$

$$\begin{aligned}\Lambda = & -\frac{v_1(z)^2\frac{\partial^2}{(\partial z)^2}U(z)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} - \frac{v_1(z)v_2(\omega)\frac{\partial^2}{(\partial z)^2}U(z)}{12\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} \\ & - \frac{v_2(\omega)^2\frac{\partial^2}{(\partial z)^2}U(z)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} - \frac{v_1(z)^2\frac{\partial^2}{(\partial \omega)^2}V(\omega)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} \\ & - \frac{v_1(z)v_2(\omega)\frac{\partial^2}{(\partial \omega)^2}V(\omega)}{12\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} - \frac{v_2(\omega)^2\frac{\partial^2}{(\partial \omega)^2}V(\omega)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} \\ & + \frac{U(z)\frac{\partial}{\partial z}v_1(z)^2}{48\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} + \frac{V(\omega)\frac{\partial}{\partial z}v_1(z)^2}{48\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} \\ & - \frac{U(z)v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} - \frac{U(z)v_2(\omega)\frac{\partial^2}{(\partial z)^2}v_1(z)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} \\ & + \frac{U(z)\frac{\partial}{\partial \omega}v_2(\omega)^2}{48\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} + \frac{V(\omega)\frac{\partial}{\partial \omega}v_2(\omega)^2}{48\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} \\ & - \frac{V(\omega)v_1(z)\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)} - \frac{V(\omega)v_2(\omega)\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(\omega) + 3v_1(z)v_2(\omega)^2 + v_2(\omega)^3\right)}\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\begin{aligned}\Psi_1 = & \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{\sqrt{U(z)}\sqrt{V(\omega)}v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & - \frac{\sqrt{U(z)}\sqrt{V(\omega)}v_2(\omega)\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} + \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial}{\partial \omega}v_2(\omega)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & - \frac{\sqrt{U(z)}\sqrt{V(\omega)}v_1(z)\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{\sqrt{U(z)}\sqrt{V(\omega)}v_2(\omega)\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)}\end{aligned}$$

$$\begin{aligned}\Psi_2 = & \frac{v_1(z)^2\frac{\partial^2}{(\partial z)^2}U(z)}{12\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} + \frac{v_1(z)v_2(\omega)\frac{\partial^2}{(\partial z)^2}U(z)}{6\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & + \frac{v_2(\omega)^2\frac{\partial^2}{(\partial z)^2}U(z)}{12\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} + \frac{v_1(z)^2\frac{\partial^2}{(\partial \omega)^2}V(\omega)}{12\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & + \frac{v_1(z)v_2(\omega)\frac{\partial^2}{(\partial \omega)^2}V(\omega)}{6\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} + \frac{v_2(\omega)^2\frac{\partial^2}{(\partial \omega)^2}V(\omega)}{12\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & - \frac{v_1(z)\frac{\partial}{\partial z}U(z)\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{v_2(\omega)\frac{\partial}{\partial z}U(z)\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & - \frac{i v_1(z)\frac{\partial}{\partial \omega}V(\omega)\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{i v_2(\omega)\frac{\partial}{\partial \omega}V(\omega)\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & + \frac{U(z)\frac{\partial}{\partial z}v_1(z)^2}{3\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{V(\omega)\frac{\partial}{\partial z}v_1(z)^2}{6\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & - \frac{U(z)v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{6\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{U(z)v_2(\omega)\frac{\partial^2}{(\partial z)^2}v_1(z)}{6\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & + \frac{i v_1(z)\frac{\partial}{\partial z}U(z)\frac{\partial}{\partial \omega}v_2(\omega)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} + \frac{i v_2(\omega)\frac{\partial}{\partial z}U(z)\frac{\partial}{\partial \omega}v_2(\omega)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\ & - \frac{v_1(z)\frac{\partial}{\partial \omega}V(\omega)\frac{\partial}{\partial \omega}v_2(\omega)}{4\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \dots\end{aligned}$$

$$\begin{aligned}
\Psi_3 = & \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{\sqrt{U(z)}\sqrt{V(\omega)}v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\
& - \frac{\sqrt{U(z)}\sqrt{V(\omega)}v_2(\omega)\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} + \frac{\sqrt{U(z)}\sqrt{V(\omega)}\frac{\partial}{\partial \omega}v_2(\omega)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} \\
& - \frac{\sqrt{U(z)}\sqrt{V(\omega)}v_1(z)\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)} - \frac{\sqrt{U(z)}\sqrt{V(\omega)}v_2(\omega)\frac{\partial^2}{(\partial \omega)^2}v_2(\omega)}{8\left(v_1(z)^3+3v_1(z)^2v_2(\omega)+3v_1(z)v_2(\omega)^2+v_2(\omega)^3\right)}
\end{aligned}$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"